THISHAKYA BANDARA

ELECTRONIC & TELECOMMUNICATION ENGINEERING

PROFILE

I am a self-taught computer architecture enthusiast passionate about **ASIC/FPGA implementation, RTL/logic design, RISC-V architecture, hardware accelerator design, and SoC development.** I am eager to contribute my skills and knowledge to your esteemed company.

EDUCATION

UNIVERSITY OF MORATUWA

June 2021 - Present

- B.Sc. (Hons.) in Electronic and Telecommunication Engineering **(Major)** and Mathematics **(Minor)**
- CGPA : 3.754/4.0 (93.85%)
- Dean's List: Semester 1, 2, 4, 6
- SystemVerilog for ASIC/FPGA Design and Simulation- Short Course By Synopsys

NALANDA COLLEGE, COLOMBO

- GCE A Level 2019 - 3 As
- Z Score - 2.301
- GCE Ordinary Level 2016- 9As

TECHNICAL STRENGTHS

Hardware Description Languages
- System Verilog, Verilog, VHDL

Digital Design and Simulation Software
- Xilinx Vivado, Vitis, Intel Quartus, Verilator

Computer Languages
- C/C++, Python, Matlab, LaTex

OS
- Linux , PetaLinux (on FPGA)

Electronic Product Design
- Altium Designer, SOLIDWORKS

Robot Simulation and Programming
- Webots, Arduino C, Embedded C

EXPERIENCES

FPGA AND DSP ENGINEER (SATCOM) - ADDVALUE TECHNOLOGIES (SINGAPORE)

- Involved in testing and developing test programs to verify and prove the functionality of the SOM board. Responsibilities included board bring-up, functional testing, debugging, validating hardware interfaces, and ensuring the board operated according to the required specifications.

EMBEDDED ENGINEER (SATCOM) - THAKSHANA TECHNOLOGIES

February - September 2025

- Involved in the design and testing of printed circuit boards (PCBs), including schematic review, component selection, layout verification, board bring-up, debugging, and functional validation.

COMPUTER ARCHITECTURE INTERN - NATIONAL UNIVERSITY OF SINGAPORE

December 2023 - June 2024

QS World University Ranking : 8th

- Designed an indexing algorithm for **vector databases targeting FPGA implementation**, a rapidly emerging field in **AI.** (NDA)
- Implemented the Limago 100GbE Ethernet framework with TCP/IP support on an FPGA stack(HACC Cluster) at NUS.

ACHIEVEMENTS

DVCON INDIA 2023 - GLOBAL PROCESSOR DESIGN AND VERIFICATION CHALLENGE - 1ST RUNNERS-UP

2023

- Competition Task : Optimizing the performance of a RISCV based **low-Power** system on Chip **(SoC)** for a **ML** application **(**designed to run on **edge devices)**.
- My Contribution : Planning improvements in the Ibex core and hardware accelerator in the SoC, RTL design, and performance evaluation.

_SRI LANKA PHYSICS OLYMPIAD

2019

- Achieved a **Silver Medal with a score of 96 out of 100.**

EXTRACURRICULAR ACTIVITIES

MEMBER OF U18 SRI LANKA NATIONAL BASEBALL SQUAD 2018

SERVED AS THE VICE-CAPTAIN OF THE U15 AND U20 BASEBALL TEAM- NALANDA COLLEGE 2014 & 2019

SRI LANKA UNIVERSITY COLOURS FOR BASEBALL (SLUSA) 2022 & 2024

3RD YEAR COLOURSMEN- NALANDA COLLEGE, COLOMBO 2016 - 2018

UNIVERSITY OF MORATUWA COLOURS FOR BASEBALL 2022- 2024

INTER-UNIVERSITY GAMES- BASEBALL CHAMPIONS 2022

SRI LANKA UNIVERSITY GAMES(SLUG)- BASEBALL CHAMPIONS 2023

INTER-UNIVERSITY- BASEBALL RUNNERS-UP 2024

BEST 2ND BASEMAN U20 SRI LANKA SCHOOL BASEBALL CHAMPIONSHIP 2018

U20 SRI LANKA SCHOOLS BASEBALL CHAMPIONS 2015 & 2017

U20 SRI LANKA SCHOOLS BASEBALL 1ST RUNNERS-UP 2018

U20 SRI LANKA SCHOOLS BASEBALL 2ND RUNNERS-UP 2016

U20 BEST BATTER U15 WESTERN PROVINCE SCHOOL BASEBALL CHAMPIONSHIP 2015

PROJECTS

OPTIMIZING PULPISSIMO SOC PLATFORM TARGETING MOBILENETV1

March 2023 - September 2023

- Pulpissimo is a **low-power SoC** designed to run **ML applications on the edge**. I contributed to researching design improvements for the Ibex core and hardware accelerator (both inside the SoC), as well as to the RTL design and performance evaluation process.

SPECTRUM ANALYZER DESIGN (FYP)

- Designed the **digital signal processing part and the control signal generation for the VCO and ADC PCBs on the FPGA**. Additionally, I tested the ADC and VCO **PCBs by integrating them with the FPGA**.
- **STM32** touch display integration to visualize data by using **QSPI** to communicate with FPGA uses **freeRTOS** for task scheduling.

MORSE CODE PRACTICE KIT

Nov 2023 – May 2024

- Developed a feature-rich Morse code practice kit, surpassing traditional devices used by the military. Enhanced functionality with a display, sound options, and additional features while maintaining affordability (under $50). Displayed at EXMO- Flagship Exhibition of the University of Moratuwa, during a telecommunication laboratory tour.
- Tools: Altium Designer, SOLIDWORKS, Arduino, C++.

REFRIGERATOR PROJECT

Nov 2023 – May 2024

- Designed an advanced refrigerator operation algorithm surpassing existing market competitors. Optimized microcontroller performance within memory constraints, utilizing existing controller board design to remain cost-effective. Employed data logging and analysis to enhance system efficiency and functionality.

CUTOFF CIRCUIT DESIGN

Nov 2023 – May 2024

- Investigated battery charging/discharging cycles, identifying flaws in the charger cutoff mechanism. Designed and implemented a compact PCB-based cutoff circuit to resolve the issue. Addressed challenging physical constraints in charger and PCB placement, carefully selecting components to fit size limitations.
- Tools: Altium Designer, SOLIDWORKS

LEAD ACID BATTERY CHARGER WITH PWM

October 2022 - January 2023

CULTIWAVE

August 2022 - June 2023

SMART SPOTTER

July 2022 - November 2022